CellFlow cytokine project — every experiment, notebook by notebook

Methods, data, what was trained, results, statistics, tables and plots · generated 2026-11-02 from the repository `/users/kdm/pavelabramau/cellflow-regen` (git: one commit, 705be78; most of the work since is uncommitted).

How to read this report. Every number is taken from a saved file or notebook output, with the source named. Where a notebook never produced a conclusion, or a job has not run, this report says so instead of filling the gap. **Only one model has ever been trained to completion in this project (notebook 06), plus two more arms in notebook 08.** Notebooks 20–22, 30, 31 are CPU-only analyses with no training. Notebooks 32, 40, 41 and the unit tests exist but had **not run** when this report was written (jobs queued).

Contents

Part	Notebook(s)	What it is	Trained a model?
0	01–05	Cluster GPU inventory, environment, dataset download, 20-cytokine subset	no
1	06	CellFlow baseline, IL-15 held out in 3 of 4 donors (donor-transfer)	**yes** (500k it)
2	07	What the learned condition embedding encodes (Part I on the model; Part II on data)	no (uses 06 model)
3	08	Family one-hot and shuffled-ESM2 controls vs ESM2	**yes** (2 more arms, 500k it each)
4	09	Receptor competence vs observed response vs the 06 model's predictions	no
5	20	90-cytokine response matrix + split-half reliability, well-level null	no
6	21	Receptor layer: binding matrix B, signs, competence R, D_binding	no
7	22 (A+B)	Representation screen; agonist/antagonist panel (Part B, new)	no
8	30, 31	Training set of all 90 cytokines; analytic baselines for every held-out ligand	no
9	32, 40, 41, tests	Stage B: arm flags, zero-shot run notebook, pilot summary	**not run yet**
10	—	Cross-cutting caveats and open issues	—

Executive summary

- **Data.** Parse Biosciences PBMC cytokine screen via `theislab/cellflow-datasets`: 9,697,974 cells × 2,000 HVGs, 12 donors, 90 cytokines + PBS, 18 cell types (13.1 GB; CC BY-NC 4.0).
- **06 — the only completed baseline.** With IL-15 seen in one donor (Donor8) and predicted for three others, CellFlow beats a 'control-mean' baseline in 3/3 donors: energy distance 16.5 vs 101.6, R^2 0.958 vs 0.821. Training took 3.84 h and peaked at 3.07 GB GPU memory.
- **08 — ESM2 content matters, but a unique ID goes a long way.** Arm A (ESM2) e-dist 16.5; B (family one-hot, IL-15 identical to seven γ-chain relatives) 66.8; C (ESM2 shuffled across cytokines, still a unique ID for IL-15) 37.9; control 101.6. The notebook's own conclusion cell was never filled in.
- **09 — receptor mRNA is necessary, not sufficient.** 7 receptor-poor ligands give no response above the well-level null; TNF-α and IGF-1 are null despite abundant receptor. The 06 model predicts near-zero for *every* unseen ligand, so it gets the poor ones right for the wrong reason and misses IFN-γ (predicted/observed = 0.23).
- **20 — half the cytokines do not visibly respond.** 46 of 90 pass split-half r ≥ 0.5; the other 44 sit at 0.84–1.13× the well-level null. The naive random-PBS-split null would have called 35 of those 44 responders. **Any ligand-vs-PBS comparison needs the well-level null** (each cytokine × donor sits in one barcoding well).
- **21 — receptor layer.** 85/90 cytokines have annotated receptors; 4 antagonists carry sign −1; 5 have no annotation.
- **22A — no representation predicts response geometry well.** ESM2 ρ = 0.046 (reliable set, Mantel p = 0.18), indistinguishable from random vectors (0.035). Signed binding ρ = 0.108 (p = 0.0009) and family one-hot ρ = 0.134 (p = 0.0003) are weak but non-zero; binding − ESM2 is +0.060 with a bootstrap CI [−0.09, 0.21] (not demonstrated).
- **22B (new) — agonist/antagonist panel.** Binding puts both antagonist pairs at the maximum distance (2.0, percentile 100) as designed. ESM2 does *not* rank IL-1Ra near IL-1β (rank 32 and 4; percentile 25 raw, 34 centred), so the strong prediction is only partly met. IL-36α/IL-36Ra are near in ESM2 (rank 2 and 2) but their response distance cannot be interpreted (split-half r 0.42 and 0.15).
- **30–31 — baselines.** The 90-cytokine training set has 3,411,214 cells. The analytic 'cytokine-mean' baseline barely improves on 'control-mean' (mean e-dist 31.3 vs 34.0; wins in 57.4% of cases) — the bar a trained model must clear is low.

- **Stage B/C.** Arm flags (`--pert_rep`, `--gate`), a receptor gate, unit tests, a papermill run notebook and a pilot array are written. **None has run; no pilot result and no PILOT.md exist.** The 72-run main experiment has not been started.

Part 0 — Notebooks 01–05: infrastructure and data

Source data (04, DATA.md)

- File `pbmc_parse.h5ad`, 13,101,592,422 bytes, sha256 c72671b3…d4c9 (HF LFS), revision 4b235c9a…; opened backed (X never loaded).
- 9,697,974 cells × 2,000 genes (all highly variable), CSR float32, 964.5 M non-zeros. X is **raw UMI integers** (max 1897, 100% integer, `uns['log1p']` empty) → every notebook normalises to CP10k then log1p. Because X holds only the 2,000 HVGs, CP10k is relative to HVG counts, not total transcripts (stated convention, kept across 07, 20, 30).
- 12 donors, 91 conditions (90 cytokines + PBS), 18 cell types, 15 cytokine families. `uns`: `esm2_embeddings` (93 entries × 2,560-d, norms 9.5–14.8, **not L2-normalised**, raw cosine distance median ≈ 0.05 → nearly collinear), `donor_embeddings` (12 × 2,000, the donor's control mean over HVGs).
- obs columns include `bc1_well`: each cytokine × donor sits in **one** barcoding well, PBS spans 6 wells per donor — the origin of the well-confound found in 09/20.

01 — GPU inventory · 02 — environment · 03 — GPU verification

- **01**: `sinfo` via `ssh max-login` on the MDC cluster: T4 (max001–010), A40 ×3 per node (max020–029), V100 and B300 nodes. No model saved, nothing trained.
- **02**: conda env `cellflow` (Python 3.11) with `cellflow-tools 0.0.9`, `jax[cuda12]`; **no rapids-singlecell**; env vars `XLA_PYTHON_CLIENT_PREALLOCATE=false`, HF/pip caches redirected off $HOME.
- **03**: on an A40 (46,068 MiB, driver 610.57, cc 8.6) JAX sees a CudaDevice; 4096² matmul works; jax 0.10.2. `cellflow.__version__` does not exist in 0.0.9 (expected AttributeError, recorded).

Environment change. Notebooks 01–09 ran on the MDC *max* cluster; from 2026-11-01 the project runs on the Cyfronet/TASK cluster (`SETUP.md`, `HARDWARE.md`): A100-80GB nodes (1 GPU, 72 GB host RAM, 42 CPUs), H100 ×4 nodes, A100 (akiv) nodes down. The new env has jax 0.10.2 + CUDA 12.9 wheels; **CUDA on the new cluster was not verified** when SETUP.md was written, and per-user QOS limits could not be read (accounting DB unreachable).

05 — The 20-cytokine subset (`pbmc_subset.h5ad`)

Deterministic recipe. **Donors (4):** Donor8 + the 3 donors with most PBS cells (Donor1 90,452; Donor12 71,179; Donor2 70,520). **Cytokines (20):** IL-15 + its 7 family members (IL-13, IL-2, IL-21, IL-4, IL-7, IL-9, TSLP) + 12 drawn with `default_rng(0)` (ADSF, BAFF, C5a, FGF-beta, GM-CSF, IFN-epsilon, IL-17B, IL-20, IL-31, IL-6, LT-alpha2-beta1, SCF) + PBS. All 18 cell types kept. Result: **1,114,081 cells × 2,000 genes** (0.33 GB, sha256 61883f33…), cells per donor Donor8 187,549 · Donor1 362,067 · Donor12 309,565 · Donor2 254,900; 275,215 PBS cells.

Part 1 — Notebook 06: CellFlow baseline (IL-15 held out in 3 of 4 donors)

Question and design

Reproduce CellFlow's PBMC tutorial (`100_pbmc`) on the 20-cytokine subset and test whether it predicts a cytokine's effect in donors where that cytokine was not observed. **Split:** IL-15 is in training for Donor8 only; IL-15 for Donor1, Donor2, Donor12 is the test set (so this is **donor-transfer**, not zero-shot ligand). PBS of all donors is available as source cells. Deviations from the tutorial: PCA on CPU (no rapids), no UMAPs, local subset file.

Data and preprocessing

- Subset from notebook 05; `normalize_total(1e4) + log1p`; PCA with `centered_pca(n_comps=100)` fitted on training data, test projected with `project_pca`.
- Embedding sanity checks: 93 ESM2 vectors of 2,560-d, not L2-normalised; all 20 treated cytokines present (PBS needs none, it is the source distribution).
- Validation subsampling: 1,000 cells per training condition, 2,000 per test condition.

What was trained

Item	Setting
Model	CellFlow, solver `otfm` (OT flow matching), deterministic condition encoder, `pooling='attention_token'`, 256-d condition embedding, output dropout 0.9
Condition inputs	perturbation covariate `cytokine_treatment` = ESM2 (2,560-d) through MLP [1024,1024] (dropout 0.5); sample covariate `donor` = donor embedding through MLP [256,256]; after pooling MLP [1024,1024]; `split_covariates=['donor']`
Vector field	time encoder [1024]×3, x-encoder [2048]×3, decoder [4096]×3, silu; concatenation conditioning; probability path constant_noise 0.5; matching `match_linear(ε=0.5, τ_a=τ_b=1)`
Optimiser	Adam 5e-5 inside `optax.MultiSteps(…, 20)` (20 micro-steps per update, so 500k iterations = 25k parameter updates)
Schedule	500,000 iterations, batch 1,024, validation every 20,000; seed 0 (CellFlow default); training cells ≈ the subset minus IL-15 in Donor1/2/12
Hardware / cost	A40 GPU (MDC). **Wall time 13,832 s = 3.84 h; peak GPU memory 3.07 GB** (JAX allocator, preallocation off)
Saved model	`models/06_baseline_CellFlow.pkl` — the only trained model object in the repository

Evaluation

Predictions for IL-15 in each test (donor, cell type) are compared with the true IL-15 cells and with the **control-mean baseline** (the donor's own PBS cells, i.e. 'no effect'). Metrics: energy distance (`compute_e_distance`) in a 30-PC space fitted on a 100k-cell subsample of the whole dataset, and R^2 (`compute_r_squared`) in gene space after PCA reconstruction; groups need ≥ 20 cells. Acceptance rule: CellFlow better than control on both metrics.

Results

donor	e-dist CellFlow	e-dist control	R^2 CellFlow	R^2 control
Donor1	7.28	55.51	0.981	0.916
Donor12	22.20	166.52	0.947	0.709
Donor2	20.09	82.70	0.946	0.838
mean	**16.53**	**101.57**	**0.958**	**0.821**

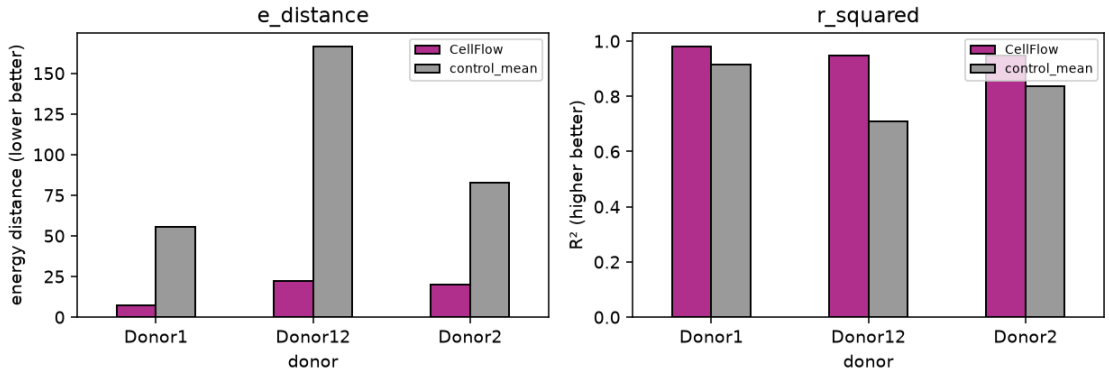


Figure 06: per-donor energy distance and R² for CellFlow vs control-mean (derived from models/06_baseline_metrics_per_donor.csv).

By cell type (mean over the 3 test donors)

cell type	e-dist CF	e-dist ctrl	R² CF	R² ctrl
B Intermediate/Memory	18.1	56.3	0.94	0.89
B Naive	18.1	49.1	0.94	0.90
CD14 Mono	28.4	232.9	0.96	0.88
CD16 Mono	67.3	235.9	0.92	0.86
CD4 Memory	22.9	129.4	0.94	0.76
CD4 Naive	17.5	144.0	0.96	0.81
CD8 Memory	77.4	236.4	0.84	0.67
CD8 Naive	35.7	153.7	0.87	0.72
MAIT	53.4	181.9	0.84	0.71
NK	34.1	149.6	0.86	0.76
NK CD56bright	47.7	208.2	0.85	0.71
Treg	46.4	125.8	0.87	0.75
cDC	35.8	151.8	0.92	0.87

- **Acceptance: PASS** — CellFlow better in 3/3 donors on both metrics. The effect is large for energy distance (6×) and moderate for R² (the control-mean baseline already has R² 0.82 in gene space, so R² has less room to separate models).
- Worst-predicted cell types by energy distance: CD8 Memory (77.4), CD16 Mono (67.3), MAIT (53.4). Best: B cells (18.1) and CD4 Naive (17.5).
- **Not tested here:** a truly unseen ligand (IL-15 was in training for Donor8); other ligands; seeds (single run, no variance estimate); any lower iteration count.

Statistics: none beyond descriptive means over 3 donors; there is no confidence interval, test or seed replicate.

Part 2 — Notebook 07: what does the learned condition embedding encode?

Question

Has CellFlow's condition encoder organised cytokines by **receptor family**, and if so did it learn that or inherit it from ESM2? Part I interrogates the 06 model; Part II asks whether the *data* (responses) have the same structure.

Part I — embeddings from the 06 model (no training)

- `cf.get_condition_embedding` on 93 cytokines × 4 donors (372 conditions, 256-d; variance is 0 since the encoder is deterministic). Spaces: ESM2 raw, ESM2 centred (mean over 93 removed), learned per donor and donor-mean.
- Statistics: cosine silhouette with `cytokine_family` labels vs **100 label permutations** (p floor 0.01); within- vs between-group cosine distance for 4 shared-receptor groups. Families with one trained member are dropped from silhouette.
- **Caveat from the notebook:** only IL-6 and IFN-epsilon of the 13 cytokines in the receptor groups were trained; the rest pass through the trained encoder zero-shot, so Part I mostly tests what the encoder does to ESM2.
- **Result (notes.md):** the learned embedding has not rediscovered receptor structure; where receptor groups are tight in it, the structure is inherited from ESM2.

Part II — do the responses have family structure? (full dataset)

Response vector per cytokine: for each donor × cell type, pseudobulk mean log1p(CP10k) minus the same donor's PBS mean (≥ 20 cells both sides), averaged over donors and concatenated over the 14 cell types with a delta for every cytokine (HSPC, ILC, Plasmablast, pDC dropped). Reliability: split-half correlation donors 1–6 vs 7–12. Cached group means: `pbmc_parse_pseudobulk_log1p_means.h5ad`.

space	14 trained cytokines (4 families) silhouette, p	77 cytokines (10 families)	40 reliable responders
response (data)	−0.021, p = 0.10	−0.118 (null −0.18), p = 0.01	−0.058 (null −0.21), p = 0.01
ESM2 raw	0.097, p = 0.04	−0.142, p = 0.01	−0.211, p = 0.04
ESM2 centred	0.139, p = 0.02	−0.020, p = 0.01	−0.080, p = 0.01
learned (donor mean)	−0.117, p = 0.17	−0.191, p = 0.01	−0.120, p = 0.01

Source: notes.md §1. A negative silhouette that still beats the permutation null (p = 0.01) means the label structure is *less negative than random*, not that clusters exist. Only the **interferons** cohere in the data (per-family silhouette +0.24); every other family is negative. ESM2 groups sequence families (complement, common β chain, IL-17); the learned space groups complement, IL-17 and IL-6/IL-12.



Figure 07: left — family silhouette (observed, purple) against 100 permuted-label draws, 77 cytokines / 10 families. Right — percentile of within-group distance vs random groups of the same size (low = tight; * ≤ 5th).

receptor group	response (data)	ESM2 raw	ESM2 centred	learned
IFNAR (IFN-α1, -β, -ε, -ω)	0.1*	0.2*	0.0*	14.4
CSF1R (M-CSF, IL-34)	21.8	9.8	3.6*	26.7
gp130 (IL-6, -11, LIF, OSM, CT-1)	84.2	0.4*	0.1*	1.7*
IL17RA/RC (IL-17A, -17F)	23.1	0.2*	0.3*	0.1*

Answer (notes.md). PBMC responses cluster by family only weakly; the single shared-receptor group tight in the data is IFNAR. gp130 and IL-17A/F are tight in ESM2 and the model but their cells do not respond alike (the

ligand-specific α chains decide a blood-cell response; IL-11 and LIF barely respond). 'Embeddings group by sequence family; cells group by the receptors they actually carry; the two agree only for IFNAR.'

Part 3 — Notebook 08: does ESM2 do real work? (family one-hot and shuffled-ESM2 controls)

Design

Identical pipeline, split, seed (0) and 500k-iteration training to notebook 06. **Only the cytokine representation changes**, and so does the first encoder layer's input size for arm B.

arm	cytokine representation	what it tests
A	ESM2 (2,560-d)	the 06 baseline; **reused from the saved 06 model** because the training PCA matched (reproducibility checked against 06's CSV: R^2 identical to 4 decimals, e-distance within 0.03 per the notes)
B	family one-hot over `cytokine_family` (10 families); IL-15 is identical to IL-2/4/7/9/13/21/TSLP	is coarse receptor-family information enough?
C	ESM2 vectors permuted across the 20 cytokine names (none keeps its own)	control: IL-15's vector is still a unique, consistent ID

What was trained

Arms B and C: two new CellFlow models, each 500k iterations, batch 1,024, same architecture and optimiser as 06 (sbatch job 3375645 per notes.md; GPU type not recorded in the files I read). Arm A: the 06 model, not retrained. **Per-arm wall time and VRAM were not extracted for this report.**

Results (cell_type = 'all', held-out IL-15)

donor	A ESM2	B family	C shuffled	control-mean
Donor1	7.3	42.1	10.6	55.5
Donor12	22.2	109.4	25.8	166.5
Donor2	20.1	48.9	77.4	82.7
mean e-dist	**16.5**	**66.8**	**37.9**	**101.6**
mean R^2	**0.958**	**0.878**	**0.921**	**0.821**
gap closed vs A, e-dist	1.00	0.41	0.75	0.00

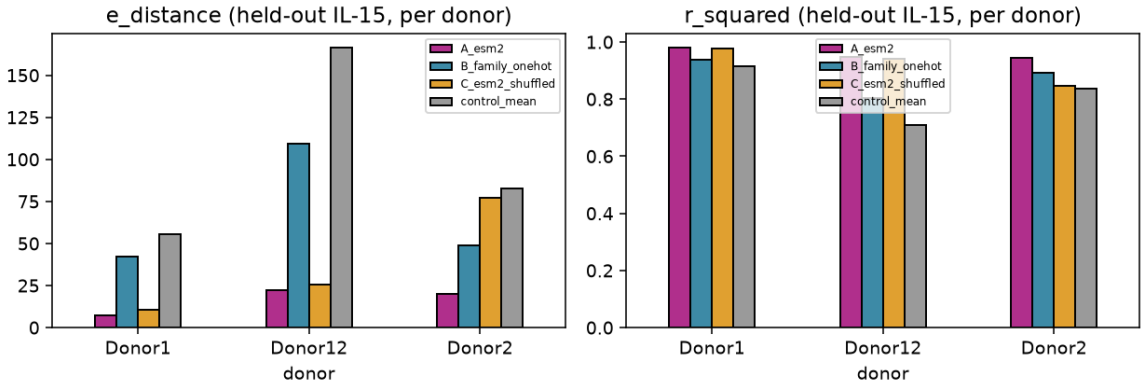


Figure 08: per-donor energy distance and R^2 for the three arms and control-mean (derived from models/08_family_vs_esm2_metrics.csv).

- **Ordering on both metrics:** A (ESM2) > C (shuffled ESM2) > B (family one-hot) > control-mean. Relative to the gap between control-mean and A, the shuffled arm closes ≈ 75% and the family one-hot ≈ 41% (e-distance, means over 3 donors).
- **Donor heterogeneity:** C ≈ A in Donor1 (10.6 vs 7.3) and Donor12 (25.8 vs 22.2) but much worse in Donor2 (77.4 vs 20.1, near control 82.7). With 3 donors and one seed the 'mean' is carried by that spread.
- **How to read it (notebook caveat):** IL-15 is in training for Donor8, so C's shuffled vector works as an ID; C ≈ A therefore says ESM2 *content* is not needed for *donor transfer*, not that the representation is ignored. B shows a generic γ-chain response is not enough. Neither speaks to unseen cytokines — that needs a split where IL-15 is absent for all donors (Stage B/C).

Status. The notebook's own 'Conclusion' cell still reads '(filled in after the run)'. The interpretation above is mine, from the saved CSVs, not the notebook's. No seeds or confidence intervals exist for this experiment.

Part 4 — Notebook 09: receptor competence vs observed response vs the 06 model

Question

Do ligands whose receptors are barely expressed on PBMCs fail to produce a response, and does the 06 model know that?

Data decisions

- **Responses:** full HF dataset (13 ligands + PBS over 12 donors in this notebook's scope; 8 of the 13 are absent from the 20-cytokine subset).
- **Receptor expression:** the HF matrix contains only 4 of 19 receptor genes, so Parse's **PBS pseudobulks** (raw counts, ~40k genes) were used: 24 files pulled out of the remote 44 GB archive by HTTP range requests (330 MB transferred) — helper `scripts/http_range_file.py`. GFRA4 is not annotated, so PSPN is scored on RET alone.
- **Cell-type mapping** from Parse labels to HF labels by name; PBS counts agree for monocytes/DC/B/NK/HSPC ($r \geq 0.9$), less for CD4 naive/memory, Treg, MAIT (0.71–0.98), poorly for ILC (0.14).

Method

- Per donor × ligand × cell type, ligand and PBS are subsampled to the same n = min(n_lig, n_PBS, 500), n ≥ 30. DE: Wilcoxon, BH FDR < 0.05, |log2FC| > 0.5. Energy distance in 50 PCs fitted on a balanced subsample. Median over donors.
- Receptor score: mean log1p(CPM) over listed subunits, averaged over donors (sensitivity: scarcest subunit).

Two method issues found (important for everything after)

null	energy distance	DE genes
random PBS halves	1.74	0.0
one PBS well vs the other PBS wells (same donor)	9.17	1.75

- **Well confound.** Each cytokine × donor is in one barcoding well; PBS spans 6. A random-halves null is too low: against it every receptor-poor ligand looks like a weak responder in 14–16 cell types; against the well null none does.
- **Energy-distance bias.** `compute_e_distance` includes the zero self-distances (a V-statistic), biased upward by ≈ (σ_X + σ_Y)/n. Floors at different n are not comparable → every comparison within a donor × cell type must use the same n.

Observed responses (well floor)

ligand	group	median DE	median e-dist	cell types > 2× well floor (e-dist)	cell types > floor (DE)
EGF	poor	2.0	9.5	0/16	0
FGF-beta	poor	2.0	10.5	1/16	0
HGF	poor	0.5	7.2	0/15	0
IGF-1	poor	3.5	9.9	1/15	1
VEGF	poor	2.0	7.0	0/15	0
GDNF	poor	1.0	6.2	0/16	0
PSPN	poor	1.0	7.4	0/15	0
GM-CSF	rich	8.0	21.8	9/17	7
IFN-γ	rich	35.0	40.8	16/16	13
IL-15	rich	150.3	165.9	16/16	16
IL-2	rich	94.0	103.1	16/16	15
IL-4	rich	46.3	55.0	14/16	14
TNF-α	rich	2.0	6.8	2/17	1

- TNF-α is effectively null despite high TNFRSF1A/1B; IGF-1 is null despite abundant IGF1R in T cells → **receptor mRNA is necessary but not sufficient**.

Receptor expression vs response (206 ligand × cell-type pairs)

pairs	Spearman ρ (DE)	Spearman ρ (e-dist)
all	0.55 (p < 1e-15)	0.53
receptor-rich only	0.34 (p = 6e-4)	0.22 (p = 0.03)
receptor-poor only	0.22 (p = 0.02)	0.24 (p = 0.01)

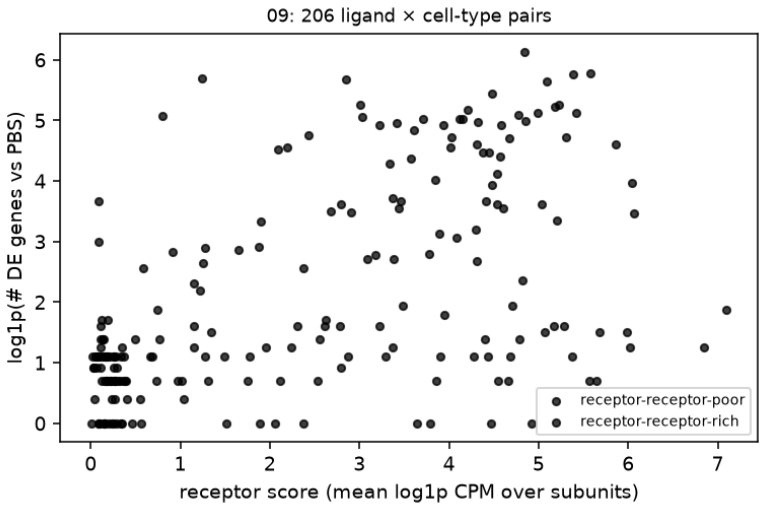


Figure 09: receptor score vs number of DE genes, coloured by group (derived from models/09_receptor_vs_response.csv). The pooled correlation is largely the two-group split.

The 06 model's predictions for these ligands (model's 100-PC space, 4 donors, matched n; well floor 21.3, random floor 13.1)

ligand	status in model	observed / floor	predicted / floor	pred / obs
EGF	unseen	1.10	0.77	0.71
FGF-beta	trained	1.11	0.79	0.71
HGF	unseen	0.92	0.73	0.79
IGF-1	unseen	1.35	0.86	0.63
VEGF	unseen	0.90	0.78	0.87
GDNF	unseen	1.01	0.90	0.90
PSPN	unseen	0.99	0.88	0.89
GM-CSF	trained	2.49	2.07	0.83
IL-15	trained (Donor8)	9.71	5.85	0.60
TNF-α	unseen	0.97	1.38	1.43
IFN-γ	**unseen**	**4.03**	**0.91**	**0.23**
IL-4	trained	5.06	6.61	1.31
IL-2	trained	5.92	4.84	0.82

Answer. The model does not predict a substantial effect for receptor-poor ligands — matching the observed null — but because it predicts a near-floor response for *every* unseen ligand, including IFN-γ (observed 4× floor). From an ESM2 vector alone it cannot separate 'receptor present' from 'receptor absent'. Receptor competence dominates whether PBMCs respond at all, and the model has no access to it. This motivates the receptor gate in Stage B.

Part 5 — Notebook 20: 90-cytokine response matrix and split-half reliability

Method

- **Source:** full dataset (9.7 M cells), CPU only. `scripts/20_pseudobulk.py` streams X once (39 s) and caches sums per (donor, cytokine, cell type, `bc1_well`) plus a seeded random 50/50 PBS split → 20,524 groups, 90 cytokines, 18 cell types.
- **Delta** (donor, cytokine, cell type) = mean over cytokine cells − mean over that donor's PBS cells of the same cell type; both sides ≥ 20 cells. 15,204 deltas of 19,440 possible. Averaged over donors that pass.
- **Response vector** = concatenation over cell types that have a delta for *every* cytokine: 14 cell types (HSPC, ILC, Plasmablast, pDC dropped; ILC missing for 47 cytokines, Plasmablast for 75) × 2,000 genes = **28,000**-d; matrix 90 × 28,000.
- **Reliability:** Pearson r between the response vectors computed from Donors 1–6 and Donors 7–12. **Null:** well-level (one PBS well vs the other PBS wells of the same donor) vs the naive random PBS split.

Results

category	count
reliable (r ≥ 0.5)	46
low r, response ≤ 2× well null ('no detectable response')	44

- All 44 low-r cytokines lie at 0.84–1.13× the well-level null (none > 1.5×) → 'no detectable response', not 'donors disagree'. Under the naive random null, 35 of the 44 would have exceeded 2× and been called responders.
- Top reliable: IFN-β 0.947, IL-15 0.923, IL-7 0.921, IL-4 0.920, IL-2 0.916, CD40L 0.915, IFN-ω 0.913, IFN-γ 0.897, IL-1β 0.878, GM-CSF 0.873. The brief expected ~40 reliable responders; this run gives 46 at r ≥ 0.5.
- **Caveat in the notebook:** r ≥ 0.5 does not prove a response — 32 of the 46 reliable cytokines (e.g. EGF, Decorin, Leptin, IGF-1 with r ≈ 0.5) are also < 1.5× the well null, so the whole-vector L2 test is blunt, and a shared well/plate-layout effect could contribute to reliability.

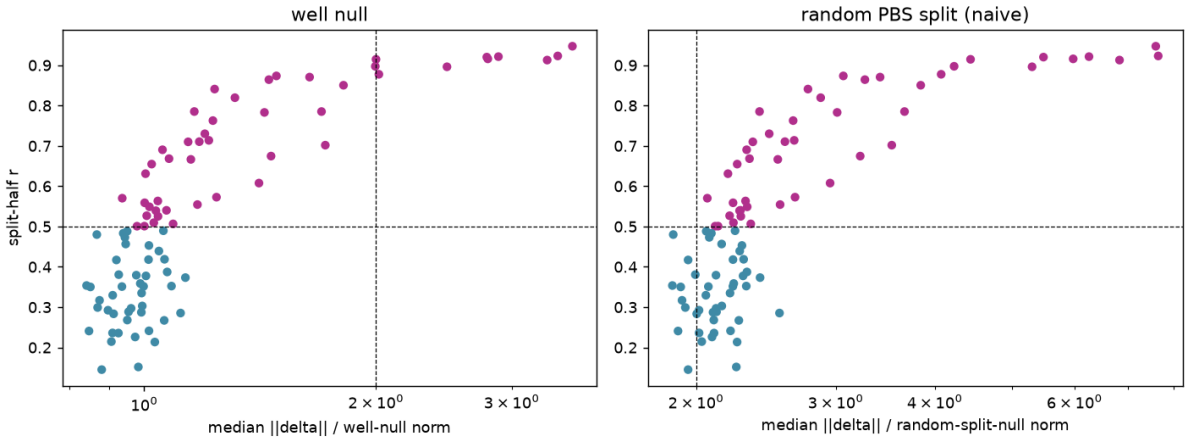


Figure 20: split-half r vs response magnitude (median ‖Δ‖) normalised to the well-level null (left) or the naive random-split null (right); purple = reliable, teal = not. Dashed: r = 0.5 and 2× null.

Reliable cytokines (all 46, split-half r)

cytokine	family	r	cytokine	family	r
IFN-beta	Interferon	0.947	IL-6	IL-6/IL-12	0.710
IL-15	Common γ chain/IL-13/TSLP	0.923	IL-1-alpha	IL-1	0.702
IL-7	Common γ chain/IL-13/TSLP	0.921	FasL	TNF	0.690
IL-4	Common γ chain/IL-13/TSLP	0.920	IL-3	Common β chain	0.675
IL-2	Common γ chain/IL-13/TSLP	0.916	Noggin	Other	0.668
CD40L	TNF	0.915	CD27L	TNF	0.666
IFN-omega	Interferon	0.913	IL-34	Growth factor	0.655
IFN-gamma	Interferon	0.897	LIGHT	TNF	0.631
IL-32-beta	Interleukin - non-classified	0.896	C5a	Complement	0.608
IL-1-beta	IL-1	0.878	IL-1Ra	IL-1	0.573
GM-CSF	Common β chain	0.873	OSM	IL-6/IL-12	0.570
IL-12	IL-6/IL-12	0.870	IGF-1	Growth factor	0.563

cytokine	family	r	cytokine	family	r
IL-21	Common γ chain/IL-13/TSLP	0.864	CD30L	TNF	0.559
IL-10	IL-10	0.850	IL-19	IL-10	0.554
TSLP	Common γ chain/IL-13/TSLP	0.841	EPO	Growth factor	0.549
TL1A	TNF	0.819	IL-24	IL-10	0.540
M-CSF	Growth factor	0.785	IL-17D	IL-17	0.539
IL-13	Common γ chain/IL-13/TSLP	0.785	Leptin	Other	0.527
IFN-alpha1	Interferon	0.783	IFN-lambda1	Interferon	0.525
IFN-epsilon	Interferon	0.763	IL-31	IL-6/IL-12	0.509
FGF-beta	Growth factor	0.730	IFN-lambda3	Interferon	0.506
CT-1	IL-6/IL-12	0.714	EGF	Growth factor	0.501
IL-27	IL-6/IL-12	0.710	Decorin	Other	0.501

Part 6 — Notebook 21: the receptor layer (binding, sign, competence)

Sources and construction

- **Binding matrix B (90 ligands × subunits):** OmniPath `LigRecExtra` (v1.0.12, academic licence). Complexes from the curated trio **CellPhoneDB, CellChatDB, ICELLNET**; origin resource kept per edge. Evidence A = ≥ 2 of the trio (83 ligands), B = 1 of the trio (Decorin), C = other OmniPath only (Noggin), none (5).
- **Sign:** CellChatDB-cofactors inhibition edges. Antagonist = −1 on its own receptor subunits and the receptors of the ligands it neutralises: **IL-1Ra, IL-36Ra, Noggin, Decorin**. (NicheNet is not used anywhere.)
- **Receptor abundance R:** Parse PBS pseudobulks, 18 archive members fetched by range requests (117 MB transferred). Per (donor, cell type) log1p(CPM).
- **Competence** per (ligand, donor, cell type): each complex is an alternative receptor; all subunits required → `competence_min` = max over complexes of the *minimum* subunit abundance (mean version = sensitivity). The mean badly overstates IL-2 competence in B naive (min 0.90 vs mean 2.86).
- **D_binding**: 90×90 distance from B, signed and unsigned. Decoys excluded (IL1R2, IL13RA2, TNFRSF11B, TNFRSF6B, TNFRSF10C/D, IL22RA2) — a curation choice flagged in `binding_edges.parquet`.

Coverage and gaps

- **85/90 annotated.** No annotation: ADSF (RETN), IL-17D, IL-32-beta, LT-alpha1-beta2, LT-alpha2-beta1 → all-zero B rows; D_binding fixes their distance to everything to 1.0 (flagged).
- Subunits per cytokine: median 2, max 16 (Noggin, inherited from its BMP/GDF targets). IFN-alpha1 annotated through alias IFNA13. GFRA4 absent from Parse annotation → **PSPN competence NaN**. Plasmablast has 11 donors.
- Assertion test passed: IL-1Ra and IL-36Ra are negative; IL-1Ra vs IL-1β signed distance 2.0 vs unsigned 0.0.

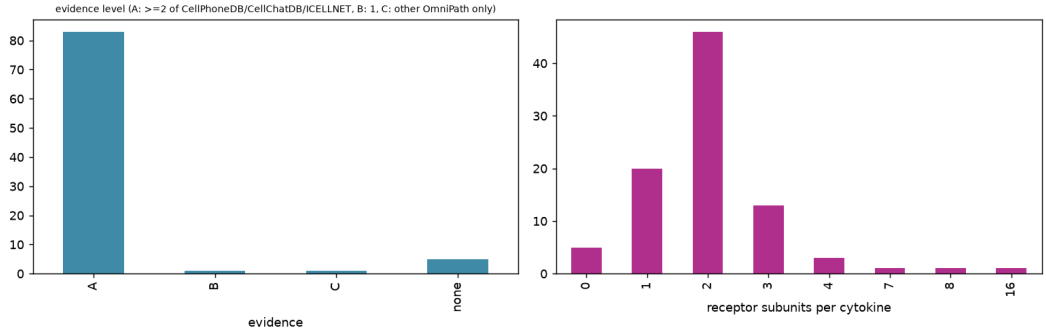


Figure 21a: evidence level (left) and number of receptor subunits per cytokine (right).

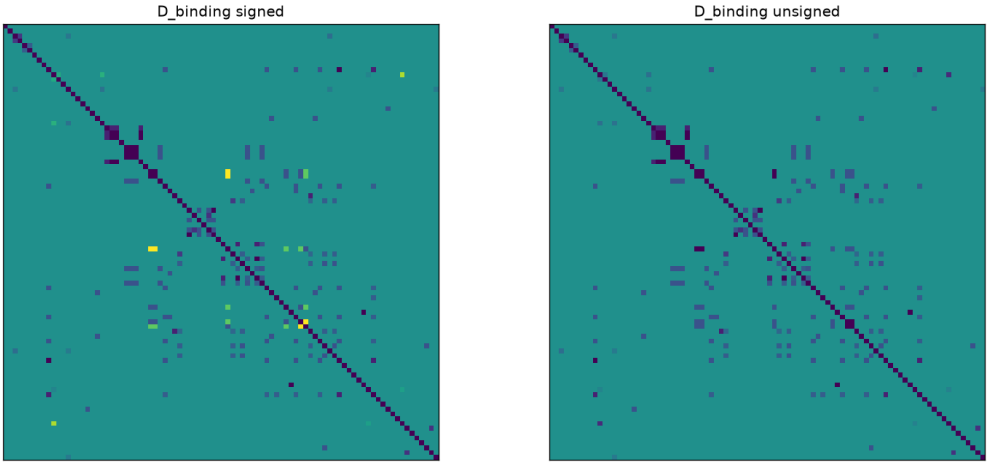


Figure 21b: signed (left) and unsigned (right) D_binding. Signed distances make antagonist–agonist pairs maximal; most of the 90×90 matrix is the constant 1.0 background.

Files: `data/binding_B.parquet, binding_B_long.parquet, binding_edges.parquet, competence_R.parquet (18,275 rows), receptor_abundance_R.parquet (215 × 123), D_binding_signed/unsigned.{npy,csv}, receptor_coverage.csv`. Licence of Parse expression data: CC BY-NC 4.0.

Part 7 — Notebook 22: representation screen (Part A) and agonist/antagonist panel (Part B)

Part A — does any condition representation predict response geometry?

Design. Each candidate representation → 90×90 cosine-distance matrix, compared with D_response (cosine distance between response vectors, 14 cell types × 2,000 genes). **Candidates:** ESM2 raw; ESM2 centred; signed binding (nb 21); family one-hot; UniProt Function text (all-MiniLM-L6-v2 sentence embeddings, mean over ≤ 200-word chunks); **constant** (control: all distances 1 → ρ undefined); **random Gaussian** 64-d (seed 0; calibration). **Subsets:** the 46 reliable responders and all 90, plus per cell type.

Statistics. (i) Spearman ρ between the flattened upper triangles. (ii) **Mantel permutation test**: candidate-matrix labels permuted jointly on rows and columns, 10,000 draws, seed 0, one-sided 'greater', p = (1+#≥obs)/(1+10,000). (iii) Distance correlation from the two distance matrices with its own label-permutation test (the V-statistic dCor is biased upward — the constant control gets 0.66 — so only its permutation p is used). **Uncorrected** p-values, 7 representations × 2 subsets.

representation	subset	ρ	Mantel p	dCor	dCor p	n
esm2_raw	reliable	0.0456	0.1754	0.494	0.0235	46
esm2_centred	reliable	0.0439	0.0964	0.482	0.0354	46
binding	reliable	0.1082	0.0009	0.637	0.0006	46
family_onehot	reliable	0.1338	0.0003	0.550	0.0004	46
constant	reliable	NaN	1.0000	0.659	1.0000	46
random_gauss	reliable	0.0354	0.1347	0.588	0.2245	46
uniprot_text	reliable	0.0642	0.0977	0.615	0.0011	46
esm2_raw	all90	-0.0170	0.7207	0.376	0.1258	90
esm2_centred	all90	0.0027	0.3684	0.363	0.1093	90
binding	all90	0.0340	0.0265	0.555	0.0006	90
family_onehot	all90	0.0668	0.0011	0.423	0.0007	90
constant	all90	NaN	1.0000	0.600	1.0000	90
random_gauss	all90	0.0243	0.0660	0.496	0.0981	90
uniprot_text	all90	-0.0083	0.6009	0.508	0.0016	90
esm2_raw	reliable	0.0400	0.2246	0.462	0.0297	46
esm2_centred	reliable	0.0406	0.1113	0.439	0.0695	46
binding	reliable	0.0924	0.0035	0.573	0.0126	46
family_onehot	reliable	0.1262	0.0002	0.510	0.0007	46
constant	reliable	NaN	1.0000	0.606	1.0000	46
random_gauss	reliable	0.0366	0.1279	0.549	0.1428	46
uniprot_text	reliable	0.0803	0.0729	0.569	0.0024	46
esm2_raw	all90	-0.0146	0.7354	0.341	0.2029	90
esm2_centred	all90	-0.0052	0.5815	0.322	0.2622	90
binding	all90	0.0268	0.0565	0.502	0.0069	90
family_onehot	all90	0.0615	0.0019	0.399	0.0007	90
constant	all90	NaN	1.0000	0.551	1.0000	90
random_gauss	all90	0.0270	0.0492	0.461	0.0752	90
uniprot_text	all90	0.0104	0.3153	0.472	0.0024	90
esm2_raw	reliable	0.0416	0.2093	0.446	0.0375	46
esm2_centred	reliable	0.0412	0.1081	0.428	0.0669	46
binding	reliable	0.0994	0.0019	0.554	0.0148	46
family_onehot	reliable	0.1353	0.0002	0.502	0.0011	46
constant	reliable	NaN	1.0000	0.584	1.0000	46
random_gauss	reliable	0.0305	0.1679	0.528	0.1721	46
uniprot_text	reliable	0.0770	0.0742	0.558	0.0013	46
esm2_raw	all90	-0.0165	0.7512	0.324	0.2133	90
esm2_centred	all90	-0.0028	0.5089	0.310	0.2207	90
binding	all90	0.0310	0.0351	0.480	0.0053	90
family_onehot	all90	0.0662	0.0010	0.385	0.0012	90
constant	all90	NaN	1.0000	0.523	1.0000	90
random_gauss	all90	0.0254	0.0583	0.438	0.0888	90
uniprot_text	all90	-0.0053	0.5717	0.450	0.0038	90

representation	subset	ρ	Mantel p	dCor	dCor p	n
esm2_raw	reliable	0.0295	0.2381	0.480	0.0317	46
esm2_centred	reliable	0.0226	0.2089	0.475	0.0337	46
binding	reliable	0.1099	0.0006	0.648	0.0001	46
family_onehot	reliable	0.1435	0.0001	0.560	0.0002	46
constant	reliable	NaN	1.0000	0.644	1.0000	46
random_gauss	reliable	0.0415	0.0969	0.579	0.1698	46
uniprot_text	reliable	0.0609	0.0869	0.620	0.0002	46
esm2_raw	all90	-0.0087	0.6109	0.356	0.1790	90
esm2_centred	all90	0.0008	0.4244	0.346	0.1382	90
binding	all90	0.0478	0.0045	0.549	0.0001	90
family_onehot	all90	0.0800	0.0002	0.426	0.0001	90
constant	all90	NaN	1.0000	0.574	1.0000	90
random_gauss	all90	0.0189	0.1233	0.470	0.2008	90
uniprot_text	all90	-0.0060	0.5780	0.497	0.0005	90
esm2_raw	reliable	0.0226	0.2821	0.470	0.0510	46
esm2_centred	reliable	0.0271	0.1736	0.467	0.0466	46
binding	reliable	0.1040	0.0012	0.643	0.0001	46
family_onehot	reliable	0.1336	0.0002	0.555	0.0001	46
constant	reliable	NaN	1.0000	0.641	1.0000	46
random_gauss	reliable	0.0381	0.1153	0.572	0.2506	46
uniprot_text	reliable	0.0600	0.0870	0.615	0.0002	46
esm2_raw	all90	-0.0027	0.5315	0.360	0.1875	90
esm2_centred	all90	0.0009	0.4171	0.352	0.1278	90
binding	all90	0.0441	0.0077	0.555	0.0001	90
family_onehot	all90	0.0809	0.0002	0.428	0.0004	90
constant	all90	NaN	1.0000	0.583	1.0000	90
random_gauss	all90	0.0143	0.1845	0.475	0.2753	90
uniprot_text	all90	0.0007	0.4713	0.503	0.0009	90
esm2_raw	reliable	0.0261	0.2764	0.413	0.1090	46
esm2_centred	reliable	0.0188	0.2397	0.391	0.1848	46
binding	reliable	0.0705	0.0182	0.523	0.0860	46
family_onehot	reliable	0.0802	0.0103	0.444	0.0243	46
constant	reliable	NaN	1.0000	0.567	1.0000	46
random_gauss	reliable	0.0227	0.2309	0.514	0.1779	46
uniprot_text	reliable	0.0257	0.2771	0.518	0.0289	46
esm2_raw	all90	-0.0111	0.6394	0.295	0.4152	90
esm2_centred	all90	-0.0077	0.6552	0.278	0.4585	90
binding	all90	0.0122	0.2210	0.443	0.0793	90
family_onehot	all90	0.0358	0.0263	0.337	0.0187	90
constant	all90	NaN	1.0000	0.496	1.0000	90
random_gauss	all90	0.0198	0.1094	0.414	0.1213	90
uniprot_text	all90	-0.0220	0.7759	0.416	0.0281	90
esm2_raw	reliable	0.0304	0.2314	0.433	0.0509	46
esm2_centred	reliable	0.0263	0.1807	0.411	0.0978	46
binding	reliable	0.1036	0.0012	0.549	0.0073	46
family_onehot	reliable	0.0947	0.0053	0.462	0.0101	46
constant	reliable	NaN	1.0000	0.570	1.0000	46
random_gauss	reliable	0.0287	0.1812	0.511	0.2483	46
uniprot_text	reliable	0.0418	0.1602	0.528	0.0136	46
esm2_raw	all90	-0.0113	0.6230	0.318	0.1636	90
esm2_centred	all90	-0.0031	0.5198	0.301	0.1956	90
binding	all90	0.0256	0.0726	0.471	0.0020	90
family_onehot	all90	0.0480	0.0098	0.363	0.0034	90
constant	all90	NaN	1.0000	0.502	1.0000	90
random_gauss	all90	0.0385	0.0117	0.433	0.0225	90
uniprot_text	all90	-0.0360	0.8740	0.430	0.0091	90

representation	subset	ρ	Mantel p	dCor	dCor p	n
esm2_raw	reliable	0.0204	0.3085	0.424	0.1000	46
esm2_centred	reliable	0.0185	0.2406	0.404	0.1638	46
binding	reliable	0.0608	0.0333	0.538	0.0791	46
family_onehot	reliable	0.0871	0.0071	0.463	0.0119	46
constant	reliable	NaN	1.0000	0.583	1.0000	46
random_gauss	reliable	0.0327	0.1470	0.533	0.1114	46
uniprot_text	reliable	0.0148	0.3506	0.529	0.0332	46
esm2_raw	all90	-0.0096	0.6218	0.312	0.3368	90
esm2_centred	all90	-0.0040	0.5428	0.295	0.3774	90
binding	all90	0.0102	0.2564	0.460	0.0833	90
family_onehot	all90	0.0369	0.0242	0.345	0.0221	90
constant	all90	NaN	1.0000	0.517	1.0000	90
random_gauss	all90	0.0212	0.0937	0.433	0.0908	90
uniprot_text	all90	-0.0242	0.8104	0.425	0.0558	90
esm2_raw	reliable	0.0338	0.2147	0.449	0.0407	46
esm2_centred	reliable	0.0337	0.1387	0.425	0.0936	46
binding	reliable	0.0944	0.0030	0.564	0.0123	46
family_onehot	reliable	0.0813	0.0101	0.468	0.0130	46
constant	reliable	NaN	1.0000	0.594	1.0000	46
random_gauss	reliable	0.0243	0.2170	0.528	0.3082	46
uniprot_text	reliable	0.0481	0.1303	0.542	0.0192	46
esm2_raw	all90	-0.0165	0.6910	0.334	0.1720	90
esm2_centred	all90	-0.0018	0.4867	0.314	0.2331	90
binding	all90	0.0163	0.1625	0.487	0.0074	90
family_onehot	all90	0.0289	0.0553	0.358	0.0134	90
constant	all90	NaN	1.0000	0.532	1.0000	90
random_gauss	all90	0.0341	0.0218	0.451	0.0408	90
uniprot_text	all90	-0.0266	0.8050	0.447	0.0142	90
esm2_raw	reliable	0.0492	0.1804	0.467	0.0341	46
esm2_centred	reliable	0.0360	0.1347	0.447	0.0728	46
binding	reliable	0.0679	0.0230	0.582	0.0172	46
family_onehot	reliable	0.0818	0.0100	0.488	0.0074	46
constant	reliable	NaN	1.0000	0.621	1.0000	46
random_gauss	reliable	0.0265	0.1985	0.559	0.1761	46
uniprot_text	reliable	0.0152	0.3630	0.561	0.0236	46
esm2_raw	all90	0.0049	0.4291	0.347	0.2657	90
esm2_centred	all90	-0.0053	0.5832	0.329	0.3015	90
binding	all90	0.0049	0.3739	0.512	0.0203	90
family_onehot	all90	0.0309	0.0473	0.373	0.0151	90
constant	all90	NaN	1.0000	0.569	1.0000	90
random_gauss	all90	0.0220	0.0857	0.474	0.0821	90
uniprot_text	all90	-0.0293	0.8041	0.470	0.0220	90
esm2_raw	reliable	0.0195	0.3226	0.435	0.0855	46
esm2_centred	reliable	0.0143	0.2811	0.410	0.1752	46
binding	reliable	0.0638	0.0281	0.554	0.0445	46
family_onehot	reliable	0.0849	0.0087	0.469	0.0123	46
constant	reliable	NaN	1.0000	0.596	1.0000	46
random_gauss	reliable	0.0325	0.1498	0.538	0.1746	46
uniprot_text	reliable	0.0173	0.3385	0.539	0.0305	46
esm2_raw	all90	-0.0189	0.7431	0.326	0.2816	90
esm2_centred	all90	-0.0054	0.5868	0.306	0.3674	90
binding	all90	0.0218	0.0966	0.484	0.0225	90
family_onehot	all90	0.0404	0.0168	0.357	0.0166	90
constant	all90	NaN	1.0000	0.535	1.0000	90
random_gauss	all90	0.0234	0.0764	0.448	0.0839	90
uniprot_text	all90	-0.0140	0.6789	0.442	0.0359	90

representation	subset	ρ	Mantel p	dCor	dCor p	n
esm2_raw	reliable	0.0167	0.3456	0.491	0.0233	46
esm2_centred	reliable	0.0327	0.1490	0.472	0.0507	46
binding	reliable	0.0804	0.0087	0.611	0.0104	46
family_onehot	reliable	0.0831	0.0082	0.506	0.0061	46
constant	reliable	NaN	1.0000	0.651	1.0000	46
random_gauss	reliable	0.0308	0.1661	0.587	0.1511	46
uniprot_text	reliable	0.0166	0.3455	0.586	0.0179	46
esm2_raw	all90	-0.0250	0.8274	0.364	0.2506	90
esm2_centred	all90	-0.0068	0.6267	0.345	0.3052	90
binding	all90	0.0244	0.0729	0.541	0.0051	90
family_onehot	all90	0.0387	0.0196	0.393	0.0087	90
constant	all90	NaN	1.0000	0.598	1.0000	90
random_gauss	all90	0.0182	0.1242	0.494	0.1119	90
uniprot_text	all90	-0.0226	0.7990	0.490	0.0280	90
esm2_raw	reliable	0.0305	0.2730	0.504	0.0323	46
esm2_centred	reliable	0.0275	0.1831	0.481	0.0776	46
binding	reliable	0.0806	0.0101	0.644	0.0048	46
family_onehot	reliable	0.1109	0.0013	0.549	0.0010	46
constant	reliable	NaN	1.0000	0.686	1.0000	46
random_gauss	reliable	0.0225	0.2378	0.614	0.1809	46
uniprot_text	reliable	0.0197	0.3428	0.618	0.0095	46
esm2_raw	all90	0.0005	0.4852	0.410	0.0834	90
esm2_centred	all90	0.0073	0.2749	0.389	0.1290	90
binding	all90	0.0221	0.0990	0.586	0.0048	90
family_onehot	all90	0.0348	0.0319	0.417	0.0091	90
constant	all90	NaN	1.0000	0.654	1.0000	90
random_gauss	all90	0.0183	0.1244	0.535	0.1510	90
uniprot_text	all90	-0.0098	0.5964	0.533	0.0167	90
esm2_raw	reliable	-0.0066	0.5182	0.451	0.0892	46
esm2_centred	reliable	-0.0027	0.4723	0.424	0.1839	46
binding	reliable	0.0636	0.0289	0.588	0.0120	46
family_onehot	reliable	0.0763	0.0149	0.477	0.0180	46
constant	reliable	NaN	1.0000	0.624	1.0000	46
random_gauss	reliable	0.0192	0.2683	0.557	0.2443	46
uniprot_text	reliable	0.0185	0.3500	0.559	0.0353	46
esm2_raw	all90	-0.0516	0.9504	0.364	0.2460	90
esm2_centred	all90	-0.0120	0.7774	0.344	0.2983	90
binding	all90	0.0214	0.1062	0.541	0.0045	90
family_onehot	all90	0.0388	0.0218	0.384	0.0195	90
constant	all90	NaN	1.0000	0.597	1.0000	90
random_gauss	all90	0.0256	0.0567	0.497	0.0631	90
uniprot_text	all90	-0.0231	0.7506	0.485	0.0442	90
esm2_raw	reliable	0.0536	0.1489	0.543	0.0020	46
esm2_centred	reliable	0.0673	0.0381	0.529	0.0038	46
binding	reliable	0.1377	0.0001	0.667	0.0001	46
family_onehot	reliable	0.1486	0.0002	0.573	0.0001	46
constant	reliable	NaN	1.0000	0.670	1.0000	46
random_gauss	reliable	0.0338	0.1426	0.593	0.3324	46
uniprot_text	reliable	0.0743	0.0850	0.638	0.0003	46
esm2_raw	all90	-0.0417	0.9359	0.428	0.0079	90
esm2_centred	all90	0.0126	0.1788	0.414	0.0097	90
binding	all90	0.0497	0.0046	0.592	0.0001	90
family_onehot	all90	0.0907	0.0001	0.460	0.0001	90
constant	all90	NaN	1.0000	0.631	1.0000	90
random_gauss	all90	0.0286	0.0395	0.520	0.1119	90
uniprot_text	all90	0.0136	0.3051	0.543	0.0001	90

Does binding beat ESM2? (object-level bootstrap of the ρ difference)

contrast	subset	mean ρ difference	95% bootstrap CI	P(diff ≤ 0)
binding - esm2_raw	reliable	+0.059	[-0.091, 0.209]	0.216
binding - esm2_centred	reliable	+0.062	[-0.082, 0.189]	0.176
binding - family_onehot	reliable	-0.027	[-0.176, 0.112]	0.645
binding - esm2_raw	all90	+0.050	[-0.034, 0.133]	0.121
binding - esm2_centred	all90	+0.031	[-0.048, 0.107]	0.216
binding - family_onehot	all90	-0.032	[-0.107, 0.039]	0.811

- **ESM2 does not predict response geometry:** ρ = 0.046 raw (p = 0.18) and 0.044 centred (p = 0.10) on reliable cytokines; −0.017 / 0.003 on all 90 — indistinguishable from random vectors (0.035, p = 0.13).
- **Only signed binding and family one-hot show signal, and it is weak** (best ρ explains ≈ 2% of rank variance). Binding − ESM2 is +0.060 (CI [−0.09, 0.21]) → directionally positive, **not demonstrated**; binding does not beat family one-hot (−0.027), so its signal may be largely the family signal.
- Per cell type (reliable set): binding Mantel p < 0.035 in all 14; family ρ highest in myeloid cells (CD14 Mono 0.143, cDC 0.149), lower in T cells (CD4 Memory 0.080, Naive 0.095). No cell-type correction.
- Caveats: 46 (not ~40) reliable responders; 5 cytokines have all-zero binding rows; IFN-ω has no UniProt Function text (embedded from its gene symbol).

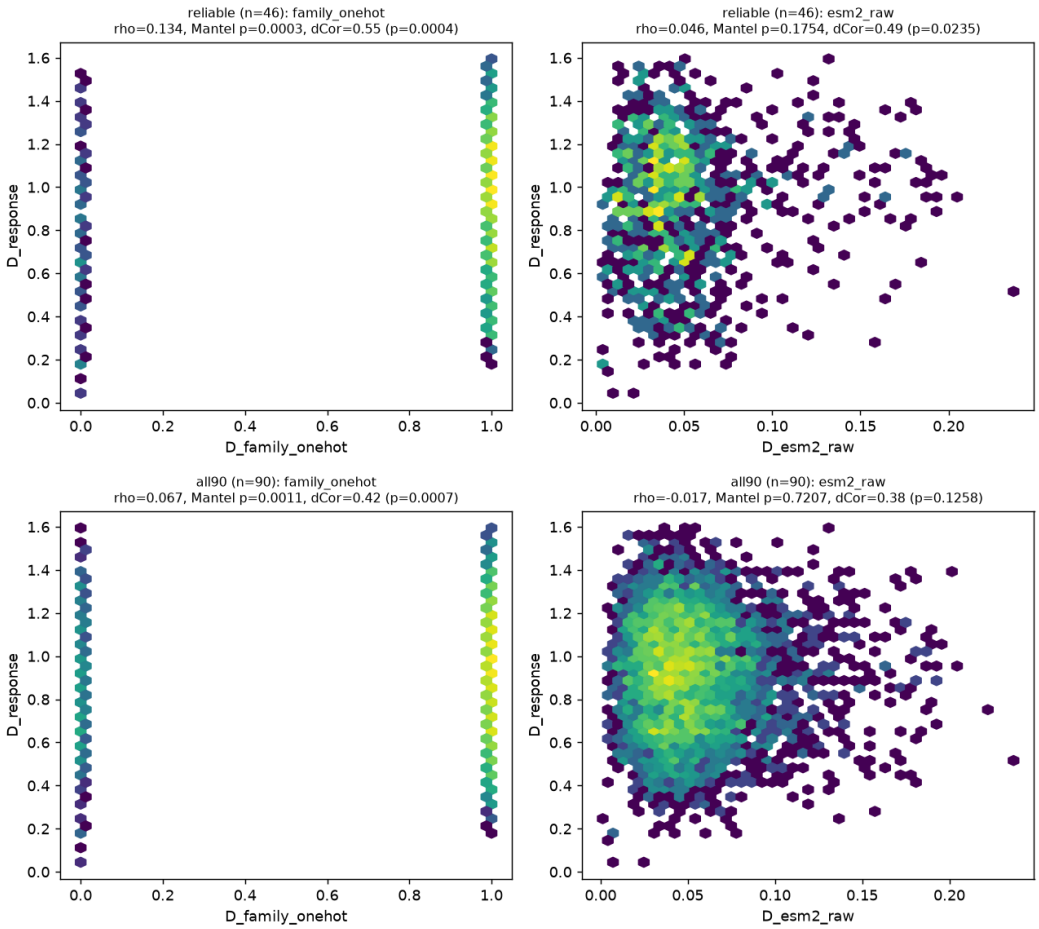


Figure 22a: D_response vs the best representation (family one-hot) and ESM2 raw, reliable (top) and all-90 (bottom). One-hot cosine distance is exactly 0 (same family) or 1 (different), hence two vertical lines.

Part B (new) — agonist / antagonist panel

Question. Does a *structural* representation (ESM2) confuse an agonist with its competitive antagonist? **Pairs:** IL-1β / IL-1Ra (IL1R1) and IL-36α / IL-36Ra (IL1RL2) as antagonist pairs; IL-2 / IL-15 (IL2RB+IL2RG) and M-CSF / IL-34 (CSF1R) as positive controls. **Stated prediction:** antagonist pairs near in ESM2, far in signed binding, far in response; controls near in all three. For each pair and space: cosine distance, rank of the partner among the 89 others (from each side; ties averaged), and percentile of the pair distance among all 4,005 pairs (0 = closest).

Reliability check first (split-half r ≥ 0.5 required for BOTH members for the response distance to be interpretable)

pair	r member A	r member B	both reliable?	interpretation
IL-1β / IL-1Ra	0.878	0.573	yes	response distance usable
IL-36α / IL-36Ra	0.419	0.152	**NO**	**both 'no detectable response'; response column uninterpretable (reported, not dropped)**
IL-2 / IL-15	0.916	0.923	yes	usable
M-CSF / IL-34	0.785	0.655	yes	usable

pair	space	distance	rank a→b	rank b→a	percentile
IL-1-beta / IL-1Ra	esm2_raw	0.034	32	4	25
	esm2_centred	0.891	38	20	34
	binding	2.000	89	88	100
	family_onehot	0.000	4	4	5
	uniprot_text	0.533	61	25	43
	random_gauss	0.799	5	8	6
	RESPONSE (observed)	1.281	66	73	92
IL-36-alpha / IL-36Ra	esm2_raw	0.019	2	2	7
	esm2_centred	0.319	2	2	2
	binding	2.000	89	89	100
	family_onehot	0.000	4	4	5
	uniprot_text	0.273	1	1	1
	random_gauss	1.064	58	60	69
	RESPONSE (observed) †	0.790	40	31	32
IL-2 / IL-15	esm2_raw	0.007	5	3	1
	esm2_centred	0.170	5	2	1
	binding	0.333	1	1	1
	family_onehot	0.000	4	4	5
	uniprot_text	0.323	11	4	3
	random_gauss	0.864	12	20	15
	RESPONSE (observed)	0.055	1	1	0
M-CSF / IL-34	esm2_raw	0.023	6	18	10
	esm2_centred	0.426	8	4	4
	binding	0.000	1	1	0
	family_onehot	0.000	6	6	5
	uniprot_text	0.310	1	1	2
	random_gauss	0.899	20	22	21
	RESPONSE (observed)	0.682	5	12	21

† response distance NOT interpretable (a member has split-half r < 0.5). 'constant' (all distances 1, percentile 50 by construction) omitted; it is in data/agonist_antagonist_panel.csv.

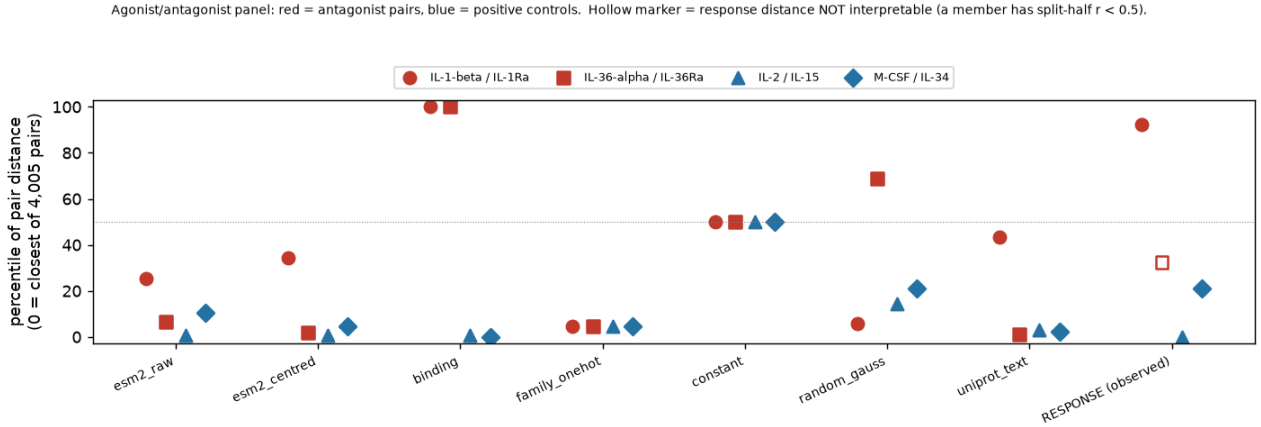


Figure 22b: percentile of pair distance in every space. Red = antagonist pairs, blue = positive controls; hollow marker = response distance not interpretable.

- **Binding behaves as designed:** both antagonist pairs sit at distance 2.0 (rank 89, percentile 100); both positive controls are at the very bottom (IL-2/IL-15 0.333 rank 1; M-CSF/IL-34 0.000 rank 1).
- **IL-1β / IL-1Ra:** ESM2 does *not* rank IL-1Ra near IL-1β — rank 32 (IL-1β's list) and 4 (IL-1Ra's list); percentile 25 raw, 34 centred. Response distance is far (1.281, percentile 92, ranks 66/73). So the response differs but ESM2 only weakly says 'similar': the strong form of the prediction ('IL-1Ra is IL-1β's nearest neighbour in

ESM2') is **not** met; the weak form (not far) is.

- **IL-36α / IL-36Ra:** ESM2 near (rank 2 and 2; percentile 7 raw, 2 centred), binding far — but the response comparison is unusable because both have no reproducible response. This pair cannot show ESM2 'failing on a functional distinction'.
- **Controls:** IL-2/IL-15 is near in every informative space (ESM2 percentile 1; response rank 1, percentile 0). M-CSF/IL-34 is near in ESM2 (percentile 10 / 4) and binding (0) and moderately near in response (0.682, rank 5/12, percentile 21).
- Family one-hot trivially puts all four pairs at 0 (percentile 5) because members share a family; random Gaussian and the constant are uninformative baselines. This is a 4-pair descriptive panel, with no statistical test.

Part 8 — Notebooks 30 and 31: training set for all 90 cytokines and analytic baselines

30 — build `data/pbmc_train90.h5ad`

- **Recipe:** 4 donors (Donor8, Donor1, Donor12, Donor2 — as in 05), **all 90 cytokines**, **cap 2,000 cells per (donor, cytokine, cell_type)** (random, seed 0), PBS in full. X = log1p(CP10k over the 2,000 HVGs); raw counts in `layers['counts']`.
- **PCA:** 100 PCs fitted on ≤ 200k random cells **excluding IL-2 and IFN-γ** (the pilot's held-out ligands), applied to all cells; explained variance of the 100 PCs = **0.497**. Stored in `obsm['X_pca']`, `uns['pca']`. Same PCA for every arm and baseline.
- Read + normalise took 17 s; file written with gzip (3.16 GB).

quantity	value
cells total	**3,411,214**
PBS cells	275,215
ligand cells	3,135,999
cells per donor	Donor1 948,373 · Donor12 946,135 · Donor2 884,495 · Donor8 632,211
(donor, cytokine, cell type) groups	6,480 possible · 6,436 non-empty · 5,278 with ≥ 20 cells · 449 at the 2,000 cap
(donor, ligand) pairs present	360 of 360
cells per ligand	min 20,856 (APRIL) · median 34,700 · max 44,469

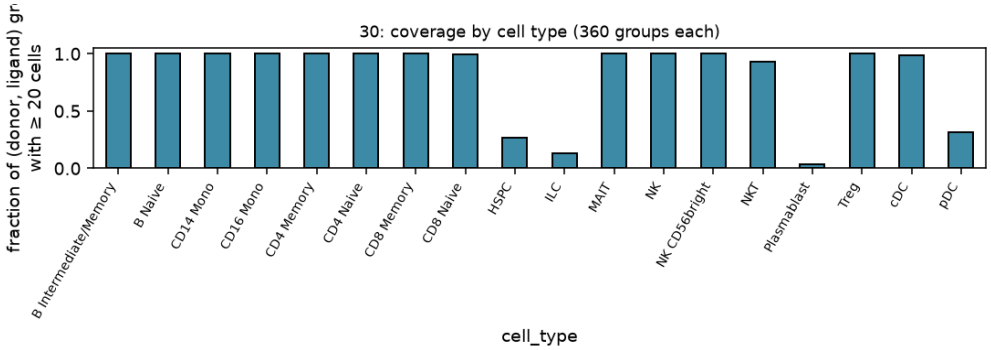


Figure 30: fraction of (donor, ligand) groups with ≥ 20 cells per cell type (derived from data/pbmc_train90_coverage.csv). HSPC 0.26, ILC 0.13, pDC 0.32, Plasmablast 0.04.

Note for the plan. The Stage B brief assumed '~700k cells'. The built training set is **3.41 M cells** (4.9× larger), so the argument that 500k iterations is 'overkill for ~700k cells' must be re-examined against the real size. The pilot (not yet run) is what would decide it.

31 — analytic baselines for every (held-out ligand, donor, cell type)

Leave-one-ligand-out over all 90 ligands, for each (donor, cell type) with ≥ 20 cells for the ligand and PBS. **control_mean** = the donor's PBS cells (no effect). **cytokine_mean** = those PBS cells translated by the mean delta of the *other 89* ligands in the same (donor, cell type) (held-out ligand never in its own mean), in PCA space for energy distance and gene space for R^2. Both sides subsampled to ≤ 1,000 cells (seed 0) so floors share n. CPU only (≈ 56 min); output `data/baselines_train90.csv` (10,554 rows, 5,277 cases). No well-level null is applied here.

baseline	mean e-distance	mean R^2
control_mean	34.04	0.9578
cytokine_mean	31.29	0.9616

cytokine_mean beats control_mean on e-distance in **57.4%** of cases and on R^2 in **65.8%** (n = 5277).

held-out ligand	control_mean e-dist	cytokine_mean e-dist	control R^2	cytokine R^2
IL-2	116.2	112.0	0.865	0.872
IFN-gamma	95.9	94.2	0.907	0.911

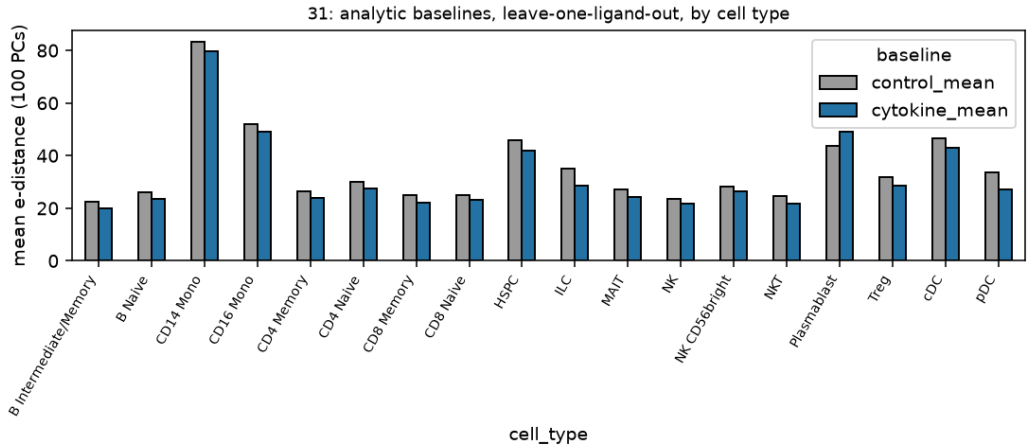


Figure 31: mean energy distance of the two analytic baselines, by cell type, over all 90 held-out ligands (derived from data/baselines_train90.csv).

Reading: the cytokine-mean baseline is only slightly better than doing nothing (a 2–4% e-distance gain for IL-2 and IFN-γ). For IL-2 and IFN-γ the distance to the truth is dominated by monocytes (CD14 Mono ≈ 220 and 406) — a model that does not move monocytes correctly cannot close the gap.

Part 9 — Stage B (flags, pilot, convergence): what exists and what has not run

Nothing in this part has produced a result. The code is written but unverified: the unit tests (job 21691739) and the slim-file build (job 21691752) were still pending in the queue when this report was generated, so no training run, no pilot, no PILOT.md, and no Stage C run exist.

Implemented (in the repository, uncommitted)

component	file	what it does
CellFlow fork	cellflow_fork/cellflow/	copy of cellflow-tools 0.0.9 edited in place (no git branches); loaded ahead of site-packages and asserted in the run notebook
arm flags	cellflow_fork/cellflow/arms.py	`--pert_rep` {constant, esm2, binding, binding_esm2, family_onehot} and `--gate` {none, receptor, receptor_permuted, mlp_full_basal}; builds the pert reps and gate features and the `prepare_data`/`prepare_model` kwargs; `split_covariates=['donor','cell_type']`
receptor gate	networks/_velocity_field.py	`CompetenceGate`: bias-free narrow MLP, output `tanh(|z|)` ∈ [0,1) multiplying the condition (ligand) latent — a mask, not a magnitude predictor; zero features ⇒ zero latent
seed fix	training/_trainer.py, model/_cellflow.py	the trainer's JAX/NumPy RNGs were hard-coded to 0, so 'seeds' would have shared one batch stream; now seeded
unit tests	tests/test_arms.py	zero competence ⇒ gated latent ≈ 0; permuting receptor features changes the output; constant pert_rep ⇒ identical ligands; tiny end-to-end train+predict for 7 arm combinations (CPU)
slim data + basal means	notebooks/32_slim_and_basal.ipynb	expression-free copy of the training file (fast loading) and PBS mean expression per (donor, cell type) for the `mlp_full_basal` gate
run notebook	notebooks/40_run_zeroshot.ipynb	papermill-parameterised: removes one ligand from ALL donors, trains, logs validation e-distance every 5k iterations with wall time, saves predictions, peak VRAM
pilot array	scripts/run_stageB_array.sbatch	8 tasks = {IL-2, IFN-γ} × {constant/none, esm2/none, binding/none, binding/receptor}, 1 seed, 200k iterations (a deliberate over-run to detect a plateau)
pilot summary	notebooks/41_pilot_summary.ipynb	wall time, peak VRAM and plateau iteration (pre-declared rule: 3-point running median; first point after which no later point improves by more than 5% (2% strict) of the total drop)

Design decisions to be aware of

- The gate multiplies the *whole* condition embedding (ligand + donor + cell type pooled), not the ligand-only latent — a simplification, because CellFlow pools all covariates before the after-pool MLP.
- Gate input for `receptor`: binding row of the ligand × receptor abundance of the (donor, cell type), 120-d. `receptor_permuted`: cell-type→abundance map shuffled within each donor. `mlp_full_basal`: binding row + full 2,000-gene PBS mean through a wider MLP — my interpretation; the brief did not define its input.
- The 'binding' pert_rep is the 120-d signed binding row of B, not the 90×90 distance matrix used in notebook 22.
- **Pending Stage C requirement:** the PCA was fitted excluding only IL-2 and IFN-γ. GM-CSF, TNF-α, IL-4 and EGF were in its fit — an unsupervised leak that should be removed (refit) or declared before the 72-run experiment.

Part 10 — Cross-cutting caveats and open issues

- **One trained baseline.** The only completed trained models are 06 and 08's arms B and C; all use a single seed, 3 test donors and a single held-out cytokine (IL-15) *that was in training for one donor*. No variance estimate exists. These do not establish zero-shot ligand generalisation.
- **Well-level null.** Any small energy distance (< 2× the well floor) must be read against the well null (09/20), which 06/08/31 do not apply. 06/08 report distances in a 30-PC space, 09 in 50 PCs, 30/31 in 100 PCs — numbers across notebooks are **not comparable**.
- **Energy-distance bias.** Comparisons within a donor × cell type must use identical n (V-statistic bias ≈ 1/n); notebooks 31 and 40 cap at 1,000 per side; 06/08 use 2,000 for validation.
- **Reliability.** Only 46/90 cytokines have a reproducible response; 44 look like pure noise at the well null, so most of the 'zero-shot to 6 ligands' targets (IL-4, EGF, TNF-α, GM-CSF...) differ greatly in how much true signal exists (EGF, TNF-α are null in 09).
- **Receptor annotation gaps.** 5 cytokines without annotation; PSPN NaN; Noggin/Decorin evidence B/C; decoy receptors excluded by choice.
- **Unfilled conclusions.** Notebook 08's conclusion cell is empty; notes.md still lists the 08 run as 'in training' although its CSVs exist; interpretation in this report is from the CSVs.
- **Infrastructure.** 01–09 ran on the MDC cluster (A40 GPUs); everything from 20 on runs on the Cyfronet cluster (CPU in practice); no GPU job has run on the new cluster yet, GPU queue is heavily contended, and the CUDA install there is unverified.
- **Version control.** Only the initial commit exists; DATA.md, HARDWARE.md, RECEPTORS.md, SETUP.md, notebooks 20–32/40/41, scripts, the fork, tests and all new data files are untracked.

File map

path	contents
notebooks/01–09, predicting_cytokines	setup, baseline, embeddings, controls, receptor competence; `predicting_cytokines` is the unmodified CellFlow tutorial stub
notebooks/20–22, 30–32, 40, 41	response matrix, receptor layer, representation screen, training set, baselines, Stage B
scripts/build/nb*.py	generators for notebooks 20–22, 30–32, 40, 41 and the verdict texts
models/	06_baseline_CellFlow.pkl, metric CSVs (06, 08, 09), figures (07, 20, 21, 22)
data/	reliability.csv, response_matrix.h5ad, binding/competence parquet, representation_screen*.csv, agonist_antagonist_panel.csv, baselines_train90.csv, pbmc_train90.h5ad (+coverage)
cellflow_fork/, tests/	Stage B code and unit tests
reports/	this report and its builder (scripts/build_report.py)